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United States Patent	12399344
Kind Code	B2
Date of Patent	August 26, 2025
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Active defocus for display assembly

Abstract

An adjustment image of a projected image is captured by a disparity image sensor. A first defocus factor is generated in response to comparing the adjustment image with a reference image driven onto a display projector assembly the generated the projected image. A second defocus factor is generated in response to an alignment of a first intensity profile of the adjustment image and a second intensity profile of the adjustment image. A focusing lens of the display projector assembly is adjusted in response to at least one of the first defocus factor and the second defocus factor.

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Appl. No.: 18/201547

Filed: May 24, 2023

Prior Publication Data

Document Identifier	Publication Date
US 20240361566 A1	Oct. 31, 2024

Related U.S. Application Data

us-provisional-application US 63462920 20230428

Publication Classification

Int. Cl.: **G02B7/02** (20210101); **G02B7/08** (20210101); **G02B27/01** (20060101); **G09G3/00** (20060101); **H04N23/55** (20230101); **H04N23/57** (20230101)

U.S. Cl.:

CPC **G02B7/028** (20130101); **G02B7/08** (20130101); **G02B27/0172** (20130101); **G09G3/002** (20130101); **H04N23/55** (20230101); **H04N23/57** (20230101); G02B2027/0138 (20130101); G02B2027/014 (20130101)

Field of Classification Search

CPC: G02B (7/028); G02B (7/08); G02B (27/0172); G02B (2027/0138); G02B (2027/014); G02B (7/09); G02B (7/282); G02B (27/0025); G09G (3/002); H04N (23/55); H04N (23/57)

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Background/Summary

CROSS-REFERENCE TO RELATED APPLICATIONS (1) This application claims priority to U.S. provisional Application No. 63/462,920 filed Apr. 28, 2023, which is hereby incorporated by reference.

TECHNICAL FIELD

(1) This disclosure relates generally to optics, and in particular to displays.

BACKGROUND INFORMATION

(2) Optical lenses are used to focus display light and also used to focus image light onto image sensors for imaging purposes. The focus of an optical lens may change with temperature. Furthermore, manufacturing tolerances of the lenses themselves or the alignment of lenses in a lens assembly may also contribute to unwanted performance variance.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

- (1) Non-limiting and non-exhaustive embodiments of the invention are described with reference to the following figures, wherein like reference numerals refer to like parts throughout the various views unless otherwise specified.
- (2) FIG. 1 illustrates a head mounted display (HMD) that may include a display assembly with an adjustable focusing lens, in accordance with aspects of the disclosure.
- (3) FIG. 2 illustrates an example display assembly having active defocus capability that may be included in an HMD, in accordance with aspects of the disclosure.
- (4) FIGS. 3A-3C illustrate example virtual images and an example adjustment image captured by a disparity image sensor, in accordance with aspects of the disclosure.
- (5) FIG. 3D-3F illustrates different intensity profiles from the dual photodiodes with respect to a focusing distance of a lens adjusted along a z-axis of the imaging system, in accordance with aspects of the disclosure.
- (6) FIG. 4 illustrates a flow chart of an example process of adjusting a focusing lens, in accordance with aspects of the disclosure.
- (7) FIG. 5 illustrates an example block diagram of an outer-loop control and an inner-loop control of adjusting a focusing lens, in accordance with aspects of the disclosure.
- (8) FIG. 6 illustrates an example block diagram of another outer-loop control for adjusting a focusing lens, in accordance with aspects of the disclosure.
- (9) FIG. 7 illustrates an example display that includes three light sources, in accordance with aspects of the disclosure.

DETAILED DESCRIPTION

(10) Embodiments of active defocusing for a display assembly are described herein. In the following description, numerous specific details are set forth to provide a thorough understanding of the embodiments. One skilled in the relevant art will recognize, however, that the techniques described herein can be practiced without one or more of the specific details, or with other methods, components, materials, etc. In other instances, well-known structures, materials, or operations are not shown or described in detail to avoid obscuring certain aspects.

(11) Reference throughout this specification to “one embodiment” or “an embodiment” means that a particular feature, structure, or characteristic described in connection with the embodiment is included in at least one embodiment of the present invention. Thus, the appearances of the phrases “in one embodiment” or “in an embodiment” in various places throughout this specification are not necessarily all referring to the same embodiment. Furthermore, the particular features, structures, or characteristics may be combined in any suitable manner in one or more embodiments.

(12) In some implementations of the disclosure, the term “near-eye” may be defined as including an element that is configured to be placed within 50 mm of an eye of a user while a near-eye device is being utilized. Therefore, a “near-eye optical element” or a “near-eye system” would include one or more elements configured to be placed within 50 mm of the eye of the user.

(13) In aspects of this disclosure, visible light may be defined as having a wavelength range of approximately 380 nm-700 nm. Non-visible light may be defined as light having wavelengths that are outside the visible light range, such as ultraviolet light and infrared light. Infrared light having a wavelength range of approximately 700 nm-1 mm includes near-infrared light. In aspects of this disclosure, near-infrared light may be defined as having a wavelength range of approximately 700 nm-1.6 μ m.

(14) In aspects of this disclosure, the term “transparent” may be defined as having greater than 90% transmission of light. In some aspects, the term “transparent” may be defined as a material having greater than 90% transmission of visible light.

(15) Display projector assemblies include focusing lenses to focus the display light. One way to increase brightness in a display projector assembly is to utilize a “faster” focusing lens with a smaller F-stop. However, these faster lenses also have a smaller depth of field that makes it more sensitive to focal length as a function of temperature. Additionally, coefficient of thermal expansion (CTE) mismatches in lens element in the focusing lens assembly may cause focal length shifts with temperatures. Furthermore, factory tolerances and manufacturing alignment tolerances may also have an impact on the system focusing performance of the system. Thus, focusing the display projector assembly with faster lenses that increase brightness may use enhanced defocusing adjustment to assist in improving the display of images.

(16) Implementations of the disclosure include adjusting a focusing lens of a display assembly in response to multiple defocus factors. A content-based defocus factor may be generated in response to compare an adjustment image (captured by an image sensor) with a reference image that was actually driven onto the display assembly. A dual-photodiode defocus factor may be generated in response to an alignment of a first intensity profile of the adjustment image and a second intensity profile of the adjustment image that is captured by the dual-photodiode disparity image sensor. These and other embodiments are described in more detail in connection with FIGS. 1-7.

(17) FIG. 1 illustrates a head mounted display (HMD) **100** that may include a display assembly with an adjustable focusing lens, in accordance with aspects of the present disclosure. HMD **100** includes frame **114** coupled to arms **111A** and **111B**. Lens assemblies **121A** and **121B** are mounted to frame **114**. Lens assemblies **121A** and **121B** may include a prescription lens matched to a particular user of HMD **100**. The illustrated HMD **100** is configured to be worn on or about a head of a wearer of HMD **100**.

(18) In the HMD **100** illustrated in FIG. 1, each lens assembly **121A/121B** includes a display waveguide **150A/150B** to direct image light generated by display projector assemblies **130A/130B** to an eyepiece region for viewing by a user of HMD **100**. Display projector assemblies **130A/130B** may include a beam-scanning display that includes a scanning mirror, for example. Display projector assemblies **130A/130B** may include one or more light sources such as a red, green, and blue light source.

(19) Lens assemblies **121A** and **121B** may appear transparent to a user to facilitate augmented reality or mixed reality to enable a user to view scene light from the environment around them while also receiving display light directed to their eye(s) by, for example, waveguides **150A/150B**.

Lens assemblies **121A** and **121B** may include two or more optical layers for different functionalities such as display, eye-tracking, face tracking, and optical power. In some embodiments, display light from display projector assemblies **130A** or **130B** is only directed into one eye of the wearer of HMD **100**. In an embodiment, both display projector assemblies **130A** and **130B** are used to direct image light into waveguides **150A** and **150B**, respectively.

(20) Frame **114** and arms **111A/111B** may include supporting hardware of HMD **100** such as processing logic **107**, wired and/or wireless data interface for sending and receiving data, graphic processors, and one or more memories for storing data and computer-executable instructions. Processing logic **107** may include circuitry, logic, instructions stored in a machine-readable storage medium, ASIC circuitry, FPGA circuitry, and/or one or more processors. In one embodiment, HMD **100** may be configured to receive wired power. In one embodiment, HMD **100** is configured to be powered by one or more batteries. In one embodiment, HMD **100** may be configured to receive wired data including video data via a wired communication channel. In one embodiment, HMD **100** is configured to receive wireless data including video data via a wireless communication channel.

(21) Processing logic **107** is illustrated as included in arm **111A** of HMD **100**, although processing logic **107** may be disposed anywhere in the frame **114** or arms **111A/111B** of HMD **100**. Processing logic **107** may be communicatively coupled to wired or wireless network **180**. Processing logic **107** may be configured to adjust a focusing lens of display projector assemblies **130A** and/or **130B** in response to defocus factors.

(22) FIG. 2 illustrates an example display assembly **200** having active defocus capability that may be included in an HMD such as HMD **100**, in accordance with aspects of the disclosure. Display assembly **200** includes two display waveguides **250A** and **250B** and a disparity image sensor **290** configured to receive a portion of the image light in display waveguide **250A** and **250B** by way of disparity waveguide **270**.

(23) An example left display projector assembly may include display **230A** and focusing lens **239A** that can be adjusted along z-axis **234A** to provide defocusing functionality to the display projector assembly. Z-axis **234A** may correspond to an optical axis of image light emitted by display **230A** and actuator **233A** is configured to adjust focusing lens **239A** along positions of z-axis **234A**. Actuator **233A** may include a micro-electro-mechanical systems (MEMS) actuator or a piezo device, for example. Processing logic **207** may be configured to drive actuator **233A** in response to one or more defocus factors. A position sensor **235A** may generate a positional signal that indicates a position of focusing lens **239A** and provide the position signal to processing logic **207** by way of communication channel X2. In the illustrated implementation, a temperature sensor **237A** is configured to generate a thermal reading of focusing lens **239A** of the display projector assembly and provide the thermal reading to processing logic **207**.

(24) In operation, an image **220A** is driven onto display **230A** and display **230A** generates image light **231A** to direct into display waveguide **250A** for presenting a virtual image to an eyebox region. The image light **231A** may be confined to propagate within display waveguide **250A** by way of total internal reflection (TIR) or otherwise. The image light is outcoupled (not specifically illustrated) from display waveguide **250A** to present a virtual image to an eyebox region. An outcoupling element (not illustrated) such a grating or a holographic optical element (HOE) may be used to outcouple the image light to the eyebox region, for example.

(25) In FIG. 2, at least a portion of the image light propagating in display waveguide **250A** is outcoupled into disparity waveguide **270**. Outcoupler **251A** outcouples the image light from display waveguide **250A** and incoupler **271A** incouples the image light into disparity waveguide **270**. The image light propagates in disparity waveguide until it is outcoupled by outcoupler **273** where camera lens assembly **280** focuses the image light to an imaging plane of disparity image sensor **290**. Disparity image sensor **290** captures an adjustment image **297** from the image light. Disparity image sensor **290** may include a complementary metal-oxide semiconductor (CMOS) image

sensor, for example.

(26) Disparity image sensor **290** also receives image light from a second display project assembly in the illustrated implementation of FIG. **2**. The second display projector assembly includes display **230B** and focusing lens **239B** and operates similarly to the first display projector assembly on the left side of FIG. **2**.

(27) Example right display projector assembly may include display **230B** and focusing lens **239B** that can be adjusted along z-axis **234B** to provide defocusing functionality to the display projector assembly. Z-axis **234B** may correspond to an optical axis of image light emitted by display **230B** and actuator **233B** is configured to adjust focusing lens **239B** along positions of z-axis **234B**. Actuator **233B** may include a micro-electro-mechanical systems (MEMS) actuator or a piezo device, for example. Processing logic **207** may be configured to drive actuator **233B** in response to one or more defocus factors. A position sensor **235B** may generate a positional signal that indicates a position of focusing lens **239B** and provide the position signal to processing logic **207** by way of communication channel X3. In the illustrated implementation, a temperature sensor **237B** is configured to generate a thermal reading of focusing lens **239B** of the right display projector assembly and provide the thermal reading to processing logic **207**.

(28) In operation, an image **220B** is driven onto display **230B** and display **230B** generates image light **231B** to direct into display waveguide **250A** for presenting a virtual image to an eyebox region. The image light **231B** may be confined to propagate within display waveguide **250B** by way of total internal reflection (TIR) or otherwise. The image light is outcoupled (not specifically illustrated) from display waveguide **250B** to present a virtual image to an eyebox region. An outcoupling element (not illustrated) such a grating or a holographic optical element (HOE) may be used to outcouple the image light to the eyebox region, for example.

(29) In FIG. **2**, at least a portion of the image light propagating in display waveguide **250B** is outcoupled into disparity waveguide **270**. Outcoupler **251B** outcouples the image light from display waveguide **250B** and incoupler **271B** incouples the image light into disparity waveguide **270**. The image light propagates in disparity waveguide until it is outcoupled by outcoupler **273** where camera lens assembly **280** focuses the image light to an imaging plane of disparity image sensor **290**.

(30) The adjustment image **297** captured by disparity image sensor **290** may include a portion of the image light **231A** from display waveguide **250A** and a portion of the image light **231B** from display waveguide **250B**.

(31) FIG. **3A** illustrates an example of a virtual image **320A** that may be driven onto display **230A** as image **220A**. FIG. **3C** illustrates an example of a virtual image **320B** that may be driven onto display **230B** as image **220B**. FIG. **3B** illustrates an example adjustment image **397** captured by disparity image sensor **290**. The example adjustment image **397** is illustrated as a slightly out of focus and blurred image of a tiger. Adjustment image **397** is an example of the adjustment image **297** that may be captured by disparity image sensor **290**. When focusing lenses **239A** and **239B** are properly focused, adjustment image **297/397** will be in focus and not blurry.

(32) Referring again to FIG. **2**, processing logic **207** is configured to receive adjustment image **297** by way of communication channel X1. Processing logic **207** may adjust focusing lens **239A** and/or **239B** in response to adjustment image **297**.

(33) In an implementation, processing logic **207** generates a content-based defocus factor in response to comparing adjustment image **297** to a reference image (e.g. image **220A** or **220B**) that was driven onto a display projector assembly to compare the expected image (the reference image) with the actual image being displayed (the adjustment image). In one implementation, generating the content-based defocus factor includes computing a relative peak signal-to-noise ratio (PSNR) between the adjustment image and the reference image and then deriving the content-based defocus factor from the relative PSNR. In one implementation, generating the content-based defocus factor includes computing a mean-square error (MSE) between the adjustment image and the reference

image and then deriving the content-based defocus factor from the MSE. In one implementation, generating the content-based defocus factor includes computing a structural similarity index measure (SSIM) between the adjustment image and the reference image and then deriving the content-based defocus factor from the SSIM. Other image quality metrics may also be generated and used to generate the content-based defocus factor.

(34) In an implementation, processing logic **207** generates a second defocus factor in response to intensity profiles from a dual-photodiode disparity image sensor that is used as disparity image sensor **290**. FIG. 2 illustrates a plan view of a section of an example image sensor **290** having Red-Green-Green-Blue (RGGB) pixels **291**. In a dual-photodiode disparity image sensor, each red subpixel includes two adjacent photodiodes configured to sense incident red light, each green subpixel includes two adjacent photodiodes configured to sense incident green light, and each blue subpixel includes two adjacent photodiodes configured to sense incident blue light. Having dual photodiodes in each subpixel allows the dual-photodiode disparity image sensor to provide defocus information from the different intensity profiles generated by the two photodiodes in the subpixels.

(35) FIGS. 3D-3F illustrate different intensity profiles from the dual photodiodes with respect to a focusing distance of a lens **339** adjusted along a z-axis **334** of the imaging system, in accordance with implementations of the disclosure. In FIG. 3D, lens **339** is positioned along z-axis **334** at a distance too close to sensor **390** and therefore does not focus light to the imaging plane of sensor **390**. Hence, the first intensity profile **381** is not aligned with the second intensity profile **382**. The first intensity profile is generated by first-photodiodes of the dual-photodiode disparity image sensor and the second intensity profile is generated by second-photodiodes of the dual-photodiode disparity image sensor disposed adjacent to the first-photodiodes.

(36) In FIG. 3E lens **339** is positioned along z-axis **334** at a distance that focuses the light to the imaging plane of sensor **390** and the first intensity profile **383** is aligned with the second intensity profile **384**. In FIG. 3F, lens **339** is positioned along z-axis **334** at a distance too far from sensor **390** and therefore does not focus light to the imaging plane. Hence, the first intensity profile **385** is not aligned with the second intensity profile **386**.

(37) While the optical path between focusing lenses **239A** and **239B** and disparity image sensor **290** in FIG. 2 is not as direct as the simplified drawings of FIG. 3D-3F, the same principle applies. Thus, the defocus of focusing lenses **239A** and **293B** can be measured in the captured adjustment image **297** by analyzing an alignment of a first intensity profile of the adjustment image and a second intensity profile of the adjustment image (where the first intensity profile is generated by first-photodiodes of the dual-photodiode disparity image sensor and the second intensity profile is generated by second-photodiodes of the dual-photodiode disparity image sensor **290**).

(38) FIG. 4 illustrates a flow chart of an example process **400** of adjusting a focusing lens, in accordance with aspects of the disclosure. The order in which some or all of the process blocks appear in process **400** should not be deemed limiting. Rather, one of ordinary skill in the art having the benefit of the present disclosure will understand that some of the process blocks may be executed in a variety of orders not illustrated, or even in parallel. All or a portion of the process blocks in FIG. 4 may be executed by processing logic **107** or **207**.

(39) In process block **405**, an adjustment image (e.g. adjustment image **297**) is captured. The adjustment image is of an image projected by a display projector assembly. The adjustment image is captured by a dual-photodiode disparity image sensor (e.g. image sensor **290** in some implementations).

(40) In process block **410**, a first defocus factor is generated in response to comparing the adjustment image with a reference image driven onto the display projector assembly.

(41) In process block **415**, a second defocus factor is generated in response to an alignment of a first intensity profile of the adjustment image and a second intensity profile of the adjustment image. The first intensity profile is generated by first-photodiodes of the dual-photodiode disparity image sensor and the second intensity profile is generated by second-photodiodes of the dual-

photodiode disparity image sensor disposed adjacent to the first-photodiodes.

(42) In process block **420**, a focusing lens (e.g. focusing lens **239A** or **239B**) is adjusted in response to the first defocus factor and the second defocus factor. In some implementations, the focusing lens is adjusted in response to only one defocus factor. By adjusting the focusing lens of the display projector assembly, the virtual image directed to the eyebox region can be more closely focused to match the actual image (e.g. **220A** or **220B**) that is driven onto the display projector assembly.

(43) In an implementation of process **400**, a third defocus factor is generated in response to a thermal reading of the focusing lens of the display projector assembly and adjusting the focusing lens of the display projector assembly is also in response to the third defocus factor. In the example of FIG. 2, thermal sensor **237A** or **237 B** may provide the thermal reading of the focusing lens to processing logic **207**.

(44) In an implementation of process **400**, the focusing lens of the display projector assembly is also adjusted in response to a position signal generated by a position sensor (e.g. **235A** or **235B**) that senses a position of the focusing lens (e.g. **239A** or **239B**) of the display projector assembly.

(45) In an implementation, generating the first defocus factor includes computing a relative peak signal to noise ratio (PSNR) between the adjustment image and the reference image and deriving the first defocus factor from the relative PSNR.

(46) FIG. 5 illustrates an example block diagram of an outer-loop control **510** and an inner-loop control **540** of adjusting a focusing lens, in accordance with aspects of the disclosure. Inner-loop control **540** includes an actuator **555** and inner-loop logic **551**. A position sensor (e.g. position sensor **235**) may provide a z-position of lens **539** to inner-loop logic **551**. Inner-loop logic **551** drives actuator **555** to adjust focusing lens **539** to a particular z-axis position in response to the position signal and a focus input from outer-loop logic **511**. Lens **539** focuses the display light emitted by display **550**.

(47) A temperature sensor **532** may sense a thermal reading **591** and provide the thermal reading **591** to thermal model **561**. Thermal model **561** may provide a defocus factor **536** to defocus module **569**. Thermal model **561** may be a linear model to generate the defocus factor **536**. The linear model may be derived from factory calibration involving monitoring defocus over multiple temperature ranges and fitting a polynomial model to the response.

(48) Disparity Image Sensor **590** receives image light from display **550** in order to capture an adjustment image. Dual photodiode defocus data **593** may be provided to Photodiode Defocus module **563** from disparity image sensor **590**. Based on intensity profiles of the dual photodiodes in disparity image sensor **590**, photodiode defocus module **563** generates defocus factor **537**.

(49) The adjustment image that disparity image sensor **590** captures may be sent to content-based analysis module **565** as frame **595**. Based on frame **595**, content-based analysis module **565** generates defocus factor **538**. Content-based analysis module **565** may also receive a reference image that was driven onto display **550** and compare the reference image with frame **595** received from disparity image sensor **590** to generate defocus factor **538**.

(50) In FIG. 5, defocus module **569** receives defocus factors **536**, **537**, and **538** and generates a defocus adjustment command **531** in response to one or more of the defocus factors **536**, **537**, and **538**. Outer-loop logic **511** receives defocus adjustment command **531** and sends it to inner-loop logic **551** as a focus input. Thus, the z-position of lens **539** is influenced by both the inner-loop control **540** and the outer-loop control **510**.

(51) FIG. 6 illustrates an example block diagram of an outer-loop control **610** for adjusting a focusing lens, in accordance with aspects of the disclosure. Outer-loop control **610** differs from outer-loop control **510** in that the defocus factor **536** from thermal model **561** is fed forward to outer-loop logic **611** and defocus module **669** receives defocus factors **537** and **538**. Defocus module **669** receives defocus factors **537** and **538** and generates a defocus adjustment command **631** in response to one or more of the defocus factors **537** and **538**. Outer-loop logic **611** generates its focus input for providing to inner-loop logic **551** in response to defocus factor **536** and defocus

adjustment command 631. The example architecture of FIG. 6 may offer improved dynamic performance as feedback from defocus factors 537 and 538 may only present smaller residual focusing errors compared to a dominant defocus estimation based on thermal modeling.

(52) FIG. 7 illustrates an example display 730 that includes three light sources 743A, 743B, and 743C, in accordance with implementations of the disclosure. Light sources 743A, 743B, and 743C may be red, green, and blue light sources, respectively. The light sources may include lasers or LEDs, for example. In some implementations of the disclosure, more than one focusing lens in a display may be adjusted along a z-axis for defocus adjustment.

(53) In FIG. 7, focusing lens 739A moves along a z-position 774A of an optical axis of light source 743A to focus illumination light 771A emitted by light source 743A. Temperature sensor 737A may generate a thermal reading of focusing lens 739A and actuator 733A is configured to adjust focusing lens 739A along z-position 774A. Similarly, focusing lens 739B moves along a z-position 774B of an optical axis of light source 743B to focus illumination light 771B emitted by light source 743B. Temperature sensor 737B may generate a thermal reading of focusing lens 739B and actuator 733B is configured to adjust focusing lens 739B along z-position 774B. And, focusing lens 739C moves along a z-position 774C of an optical axis of light source 743C to focus illumination light 771C emitted by light source 743C. Temperature sensor 737C may generate a thermal reading of focusing lens 739C and actuator 733C is configured to adjust focusing lens 739C along z-position 774C. Thus, defocusing adjustments may be made for each light source (and each corresponding wavelength) in a display 730.

(54) Embodiments of the invention may include or be implemented in conjunction with an artificial reality system. Artificial reality is a form of reality that has been adjusted in some manner before presentation to a user, which may include, e.g., a virtual reality (VR), an augmented reality (AR), a mixed reality (MR), a hybrid reality, or some combination and/or derivatives thereof. Artificial reality content may include completely generated content or generated content combined with captured (e.g., real-world) content. The artificial reality content may include video, audio, haptic feedback, or some combination thereof, and any of which may be presented in a single channel or in multiple channels (such as stereo video that produces a three-dimensional effect to the viewer). Additionally, in some embodiments, artificial reality may also be associated with applications, products, accessories, services, or some combination thereof, that are used to, e.g., create content in an artificial reality and/or are otherwise used in (e.g., perform activities in) an artificial reality. The artificial reality system that provides the artificial reality content may be implemented on various platforms, including a head-mounted display (HMD) connected to a host computer system, a standalone HMD, a mobile device or computing system, or any other hardware platform capable of providing artificial reality content to one or more viewers.

(55) The term “processing logic” in this disclosure may include one or more processors, microprocessors, multi-core processors, Application-specific integrated circuits (ASIC), and/or Field Programmable Gate Arrays (FPGAs) to execute operations disclosed herein. In some embodiments, memories (not illustrated) are integrated into the processing logic to store instructions to execute operations and/or store data. Processing logic may also include analog or digital circuitry to perform the operations in accordance with embodiments of the disclosure.

(56) A “memory” or “memories” described in this disclosure may include one or more volatile or non-volatile memory architectures. The “memory” or “memories” may be removable and non-removable media implemented in any method or technology for storage of information such as computer-readable instructions, data structures, program modules, or other data. Example memory technologies may include RAM, ROM, EEPROM, flash memory, CD-ROM, digital versatile disks (DVD), high-definition multimedia/data storage disks, or other optical storage, magnetic cassettes, magnetic tape, magnetic disk storage or other magnetic storage devices, or any other non-transmission medium that can be used to store information for access by a computing device.

(57) Networks may include any network or network system such as, but not limited to, the

following: a peer-to-peer network; a Local Area Network (LAN); a Wide Area Network (WAN); a public network, such as the Internet; a private network; a cellular network; a wireless network; a wired network; a wireless and wired combination network; and a satellite network.

(58) Communication channels may include or be routed through one or more wired or wireless communication utilizing IEEE 802.11 protocols, short-range wireless protocols, SPI (Serial Peripheral Interface), I.sup.2C (Inter-Integrated Circuit), USB (Universal Serial Port), CAN (Controller Area Network), cellular data protocols (e.g. 3G, 4G, LTE, 5G), optical communication networks, Internet Service Providers (ISPs), a peer-to-peer network, a Local Area Network (LAN), a Wide Area Network (WAN), a public network (e.g. "the Internet"), a private network, a satellite network, or otherwise.

(59) A computing device may include a desktop computer, a laptop computer, a tablet, a phablet, a smartphone, a feature phone, a server computer, or otherwise. A server computer may be located remotely in a data center or be stored locally.

(60) The processes explained above are described in terms of computer software and hardware. The techniques described may constitute machine-executable instructions embodied within a tangible or non-transitory machine (e.g., computer) readable storage medium, that when executed by a machine will cause the machine to perform the operations described. Additionally, the processes may be embodied within hardware, such as an application specific integrated circuit ("ASIC") or otherwise.

(61) A tangible non-transitory machine-readable storage medium includes any mechanism that provides (i.e., stores) information in a form accessible by a machine (e.g., a computer, network device, personal digital assistant, manufacturing tool, any device with a set of one or more processors, etc.). For example, a machine-readable storage medium includes recordable/non-recordable media (e.g., read only memory (ROM), random access memory (RAM), magnetic disk storage media, optical storage media, flash memory devices, etc.).

(62) The above description of illustrated embodiments of the invention, including what is described in the Abstract, is not intended to be exhaustive or to limit the invention to the precise forms disclosed. While specific embodiments of, and examples for, the invention are described herein for illustrative purposes, various modifications are possible within the scope of the invention, as those skilled in the relevant art will recognize.

(63) These modifications can be made to the invention in light of the above detailed description. The terms used in the following claims should not be construed to limit the invention to the specific embodiments disclosed in the specification. Rather, the scope of the invention is to be determined entirely by the following claims, which are to be construed in accordance with established doctrines of claim interpretation.

Claims

1. A head mounted display comprising: a display projector assembly configured to present a virtual image to an eyebox region, wherein the display projector assembly includes a focusing lens configured to focus the virtual image to the eyebox region; a disparity waveguide configured to receive a portion of image light of the virtual image generated by the display projector assembly; a dual-photodiode disparity image sensor configured to capture an adjustment image from the image light propagating in the disparity waveguide; and processing logic configured to: generate a first defocus factor in response to comparing the adjustment image with a reference image driven onto the display projector assembly; generate a second defocus factor in response to an alignment of a first intensity profile of the adjustment image and a second intensity profile of the adjustment image; and adjust the focusing lens of the display projector assembly in response to (1) the first defocus factor; and (2) the second defocus factor.

2. The head mounted display of claim 1 further comprising: a temperature sensor configured to

generate a thermal reading of the focusing lens of the display projector assembly, wherein the processing logic is further configured to: generate a third defocus factor in response to thermal reading; and adjust the focusing lens in response to the third defocus factor.

3. The head mounted display of claim 1, wherein the adjusting the focusing lens of the display projector assembly is also in response to a position signal generated by a position sensor that senses a position of the focusing lens of the display projector assembly.

4. The head mounted display of claim 1, wherein the display projector assembly includes a first light source, a second light source, and a third light source, and wherein adjusting the focusing lens includes: adjusting a first focusing lens along a first optical axis of the first light source, wherein the first focusing lens receives first illumination light from the first light source; adjusting a second focusing lens along a second optical axis of the second light source, wherein the second focusing lens receives second illumination light from the second light source; and adjusting a third focusing lens along a third optical axis of the third light source, wherein the third focusing lens receives third illumination light from the third light source, the first illumination light, the second illumination light, and the third illumination light having different wavelengths.

5. The head mounted display of claim 4, wherein the first illumination light is red, the second illumination light is green, and the third illumination light is blue.

6. The head mounted display of claim 1, wherein generating the first defocus factor includes: computing a relative peak signal to noise ratio (PSNR) between the adjustment image and the reference image; and deriving the first defocus factor from the relative PSNR.

7. The head mounted display of claim 1, wherein adjusting the focusing lens of the display projector assembly includes adjusting the focusing lens along a z-axis with respect to a display of the display projector assembly.

8. The head mounted display of claim 1 further comprising: a camera lens configured disposed between the disparity waveguide and the dual-photodiode disparity image sensor, wherein the camera lens is configured to focus the portion of image light of the virtual image onto an imaging plane of the dual-photodiode disparity image sensor.

9. The head mounted display of claim 1 further comprising: a display waveguide configured to receive the image light of the virtual image from the display projector assembly, wherein the disparity waveguide is configured to receive the portion of the image light of the virtual image from the display waveguide.

10. The head mounted display of claim 1, wherein the first intensity profile is generated by first-photodiodes of the dual-photodiode disparity image sensor, and wherein the second intensity profile is generated by second-photodiodes of the dual-photodiode disparity image sensor disposed adjacent to the first-photodiodes.

11. A display assembly comprising: a display; a focusing lens configured to focus display light generated by the display; a dual-photodiode disparity image sensor configured to capture an adjustment image from the display light; and processing logic configured to: generate a first defocus factor in response to comparing the adjustment image with a reference image driven onto the display; generate a second defocus factor in response to an alignment of a first intensity profile of the adjustment image and a second intensity profile of the adjustment image; and adjust the focusing lens in response to (1) the first defocus factor; and (2) the second defocus factor.

12. The display assembly of claim 11 further comprising: a temperature sensor configured to generate a thermal reading of the focusing lens, wherein the processing logic is further configured to: generate a third defocus factor in response to thermal reading; and adjust the focusing lens in response to the third defocus factor.

13. The display assembly of claim 11, wherein the adjusting the focusing lens is also in response to a position signal generated by a position sensor that senses a position of the focusing lens.

14. A method comprising: capturing an adjustment image of an image projected by a display projector assembly, wherein the adjustment image is captured by a dual-photodiode disparity image

sensor; generating a first defocus factor in response to comparing the adjustment image with a reference image driven onto the display projector assembly; generating a second defocus factor in response to an alignment of a first intensity profile of the adjustment image and a second intensity profile of the adjustment image, wherein the first intensity profile is generated by first-photodiodes of the dual-photodiode disparity image sensor, and wherein the second intensity profile is generated by second-photodiodes of the dual-photodiode disparity image sensor disposed adjacent to the first-photodiodes; and adjusting a focusing lens of the display projector assembly in response to the first defocus factor or the second defocus factor.

15. The method of claim 14 further comprising: generating a third defocus factor in response to a thermal reading of the focusing lens of the display projector assembly, wherein the adjusting the focusing lens of the display projector assembly is also in response to the third defocus factor.

16. The method of claim 14, wherein the adjusting the focusing lens of the display projector assembly is also in response to a position signal generated by a position sensor that senses a position of the focusing lens of the display projector assembly.

17. The method of claim 14, wherein adjusting the focusing lens of a display projector includes: adjusting a first focusing lens along a first optical axis of a first light source, wherein the first focusing lens receives first illumination light from the first light source; adjusting a second focusing lens along a second optical axis of a second light source, wherein the second focusing lens receives second illumination light from the second light source; and adjusting a third focusing lens along a third optical axis of a third light source, wherein the third focusing lens receives third illumination light from the third light source, the first illumination light, the second illumination light, and the third illumination light having different wavelengths.

18. The method of claim 17, wherein the first illumination light is red, the second illumination light is green, and the third illumination light is blue.

19. The method of claim 14, wherein generating the first defocus factor includes: computing a relative peak signal to noise ratio (PSNR) between the adjustment image and the reference image; and deriving the first defocus factor from the relative PSNR.

20. The method of claim 14, wherein adjusting the focusing lens of the display projector assembly includes adjusting the focusing lens along a z-axis with respect to a display of the display projector assembly.
